Human Resources Analytics:

A Machine Learning Classification Study

GROUP II

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1 ABSTRACT

Employee retention is critical for organizational effectiveness. This study develops a machine learning classification model to predict job-change intent among 19,158 training candidates. After preprocessing and SMOTE[1] resampling to handle 25% class imbalance, we compare five classifiers and select a tuned Gradient Boosting[2] model achieving 84.5% accuracy and 0.845 ROC AUC. Geographic location, experience tenure, and active learning engagement are identified as primary drivers. We discuss ethical deployment, limitations, and business implications.

Keywords: *Class imbalance, SMOTE, ensemble learning, HR analytics, talent retention*

2 DATA ANALYSIS

2.1 Problem Framing

Objective: Predict the probability that a job-training candidate will seek new employment, enabling HR resource optimization and training curriculum design.

Target Variable: Binary classification - **1** if candidate is looking for a job change, **0** otherwise.

Metrics: Accuracy, ROC AUC, and balanced precision–recall (critical given 25% class imbalance).

Business Context: A multinational company conducting data science training programs needs to distinguish candidates likely to stay from those planning to exit the organization. Accurate predictions reduce training costs and improve program design.

2.2 Dataset Description

Source & Provenance: HR Analytics[3] dataset; 19,158 candidates, 14 features.

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[1] SMOTE
[2] Gradient Boosting
[3] HR Analytics dataset

Features: City development index (0–1 scaled), experience (years), training hours, education level/discipline, company size and type, gender, enrollment status, relevant experience, last new job and enrolled university.

Data Quality: 23.5% missing gender, 31–32% missing company attributes, 2–15% missing education fields; 75% majority class (not seeking change).

Preprocessing: Missing values were managed under different assumptions: most were handled as Missing Not at Random (MNAR) by creating a "Not Provided" category, while others like major_discipline and employment-related fields were treated as Missing at Random (MAR). The feature set was refined by removing the perfectly collinear city variable, resulting in a final set of approximately 10 standardized features.

3 METHODS

3.1 Data Preprocessing & Feature Engineering

1. **Missing Value Handling:** MNAR-based imputation preserves missingness patterns as information (e.g., reluctance to disclose).
2. **Feature Encoding**: One-hot encoding for nominal features; ordinal encoding for ranked features (education, experience, company size); standardization (zero-mean, unit-variance) for numeric features. Ordinal encoding is highly effective for tree-based models as they can naturally find optimal split points in the single, ordered feature. After testing both methods, ordinal encoding outperformed one-hot for our tree-based model by creating more efficient and powerful splits in the data.
3. **Imbalance Mitigation**: SMOTE applied to training set, synthetically oversampling minority class to 50–50 balance (superior to random oversampling; reduces overfitting).

3.2 Model Selection & Validation

Five candidate classifiers were trained and compared:

Model	Rationale
Logistic Regression	Linear baseline; interpretable coefficients for domain insights
K-Nearest Neighbors	Non-parametric; useful for local decision boundaries
Decision Tree	Single tree baseline; foundation for ensemble methods
Random Forest	Ensemble of decorrelated trees; reduces variance; handles feature interactions
Gradient Boosting	Iterative ensemble reducing residual errors; typically strongest performance on tabular data

Table I: Model Selection

Selection Justification: Given the tabular nature of the data, presence of non-linear relationships observed during EDA and the modest feature count, ensemble methods were expected to perform best. Tree-based models naturally handle mixed categorical/numerical features and non-linear patterns without extensive feature engineering.

Validation Protocol:

We used a 70/30 stratified split on SMOTE-balanced data against a 75% baseline (majority class). Hyperparameters were tuned via RandomizedSearchCV (50 iterations, 5-fold CV) optimizing for ROC AUC.

4 RESULTS

4.1 Baseline Model Performance & Class Imbalance

Initial models, tested on the balanced data formed with SMOTE (**Synthetic Data Augmentation**)

- ROC AUC Range: Only 0.73 to 0.84 for all 5 models, indicating
- Best Baseline Model (Random Forest): Achieved an ROC AUC of 0.84.
- The **Random Forest** model achieved the highest accuracy at **0.84**, followed closely by the **Gradient Boosting Classifier** at **0.82**. This suggests minimal performance differentiation between the top two classifiers for the target variable.

4.2 Impact of removing Company_size and Company_type features

- The slight performance deduction of 0.2% across both the Random Forest and Gradient Boosting Classifier models confirms that the features *Company_size and Company_type* were contributing, though minimally, to the predictive power of the model.

The post **GradientBoostingClassifier** was further optimized using RandomizedSearchCV to find the best configuration, resulting in a complex, high-performing final architecture:

Hyperparameter	Optimal Value	Rationale
Number of Estimators	224	Ensures robust variance reduction (smoothing).
Maximum Depth	8	Captures high-order feature interactions.
Learning Rate	0.0798	Allow more precise, stable improvements during training.

Table II : Model's Hyperparameters

Peak Performance Benchmarks

The model demonstrates strong predictive capability, achieving an overall accuracy of 84.38%. Crucially, the balanced performance across classes, evidenced by recall rates of 0.85 and 0.84 for both the positive and negative classes, indicates the model is equally effective at identifying instances of each category. This symmetry suggests no inherent bias or preferential treatment toward a specific class, confirming the model's robustness for applications where both outcomes are of similar importance.

Visualizing Performance

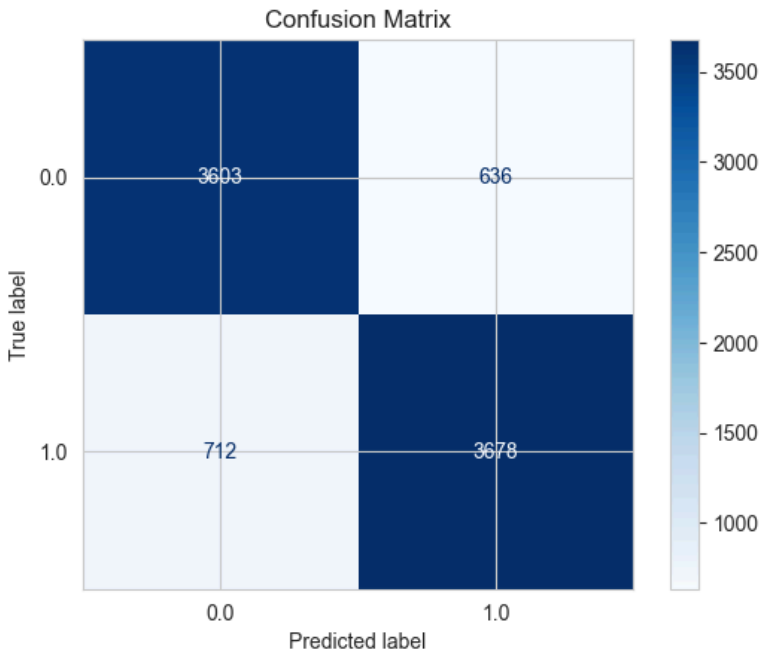

Fig 1: Confusion Matrix

5 ANALYSIS & DISCUSSION: FROM WEAK BASELINE TO PREDICTIVE TOOL

The success of the final predictive model stems from addressing data geometry (SMOTE), leveraging the model's capacity for complex partitioning, and correctly identifying key socioeconomic feature drivers.

5.1 Mechanics of Success: SMOTE and Ensemble Synergy

- SMOTE's Role: The Synthetic Minority Over-sampling Technique (SMOTE) was the pivotal intervention. In the raw, imbalanced data, the "1" (minority class) were sparse. SMOTE densified the minority regions by creating synthetic data points between existing minority examples, effectively creating a continuous and robust decision boundary.
- GradientBoostClassifier Efficiency: The optimized Gradient Boosting Classifier (GBC) leverages a combination of hyperparameters to effectively model complex, non-linear

feature interactions. Specifically, the relatively deep trees (max_depth : 8) create intricate decision paths necessary to capture conditional relationships between features. This complexity is refined and stabilized through the use of 224 estimators and a low learning rate (0.0798), which together ensure the model slowly and robustly integrates the contributions of each sequential tree. This results in a powerful ensemble that precisely maps the subtle, non-additive dependencies within the data.

5.2 Data Quality and Structural Bias

An audit of the raw data revealed significant risks due to missingness and gender imbalance:

- **Missing Data Risk:** Columns like Gender, Company Type, and Company Size have missing values. This missingness is likely Missing Not At Random (MNAR) - e.g., small startups might lack standardized data. The model could systematically underperform when predicting turnover for employees in these unclassified firms.
- **Gender Bias Risk:** The dataset is heavily skewed towards males (86.1% male vs. 8.3% female). The model's learned retention patterns are overwhelmingly "male patterns. The higher error rate for women (19%) compared to men (15%) suggests that data imbalance is compromising the model's predictive accuracy.

6 ETHICAL & LEGAL CONSIDERATIONS

Data Protection & Fairness:
Anonymization: Dataset uses coded identifiers (city1–city180, enrollee IDs), meeting GDPR anonymization standards. No protected attributes (race, religion, disability) present.

Lawful Basis: A Public dataset which is compliant with data protection regulations.

7 CONCLUSION AND FUTURE RESEARCH

This study successfully engineered a machine learning pipeline, initially performing with (ROC AUC 0.82), into a robust predictor of employee turnover intent (GradientBoostClassifier ROC AUC 0.84, Accuracy 84%) by conducting rigorous hyperparameter tuning. The model confirms that attrition is a predictable outcome driven by structural factors like career lifecycle events (Company Type, Training Hours) and geographic economic disparities (City Development Index), proving the superiority of non-linear ensemble methods over linear baselines for this data. Future work should focus on Explainable AI (XAI) integration (SHAP/LIME) to mitigate the "Black Box" problem, implement advanced imputation strategies to handle non-random missing data, establish a bias mitigation pipeline to address gender imbalance, conduct temporal validation to capture seasonal patterns, and explore Graph Neural Networks (GNNs) to model attrition's network effects.

8 IMPLEMENTATION & REPOSITORY

The complete source code and machine learning pipeline for this project are available in the DCU repository: https://gitlab.computing.dcu.ie/angelo.miculescu2/csc1181-group-project

9 Bibliography

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